

SHAYAN K DARIAN

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EDUCATION

California State University, Fullerton, Fullerton, California

August 2022 - May 2025

Master of Science, Computer Science

- GPA: 3.97/4.00

University of California, Los Angeles, Los Angeles, California

September 2016 - September 2019

Bachelor of Science, Electrical Engineering

Technical Breadth in Technology Management & Engineering Science

WORK EXPERIENCE

California State University, Fullerton, Fullerton, California

January 2025 - May 2025

Teaching Associate

- Taught CPSC 335 Algorithms Engineering to a class of 3rd and 4th year undergraduate level students

Infosys, Los Angeles, California

October 2019 - January 2023

Technology Analyst (July 2022 - January 2023)

Senior Associate Software Engineer (November 2020 - June 2022)

Associate Software Engineer (October 2019 - October 2020)

- Implemented RESTful API servers for e-commerce applications
- Efficiently designed user-interactive pages using HTML5, CSS3, Bootstrap, JavaScript, Angular 8, and Node JS
- Created server-side applications to communicate with the database using Spring Boot and Hibernate
- Served as a consultant for Southern California Edison on an Agile ArcGIS team, responsible for application, server, and database maintenance, as well as troubleshooting software-related issues impacting field engineers' devices

PROJECTS

[Enhanced AI Movie Recommender System](#)

January 2025 - May 2025

- Engineered a hybrid movie recommender system in Python, combining content-based and collaborative-based filtering to address the overspecialization and cold start limitations in traditional collaborative-based models
- Optimized algorithm selection and improved recommendation accuracy by refining data preprocessing, feature engineering, and similarity metrics using machine learning libraries such as Pandas, NumPy, and Scikit-Learn
- Designed the system to scale with larger datasets, reducing runtime by implementing efficient vectorization techniques

[Pokémon Battle Arena](#)

April 2025 - May 2025

- Developed a decentralized Pokémon game on the Ethereum blockchain in Solidity, using Hardhat and React
- Designed NFT-based Pokémon generation with unique attributes, implementing a battle system, stat progression, leaderboard, and experience-based leveling mechanics
- Integrated blockchain-based battle history storage to ensure transparent, verifiable gameplay records

[Rate My Professors Sentiment Analysis](#)

April 2024 - May 2024

- Built a Transformer-based neural network in PyTorch to predict quality and difficulty scores from professor reviews, achieving higher accuracy than an equivalent recurrent neural network (RNN) model
- Performed extensive model evaluation and fine-tuned hyperparameters to improve prediction accuracy
- Created data pipelines for text preprocessing, tokenization, and embedding generation to improve model interpretability and robustness

[EMNIST Image Recognition & Generation](#)

March 2024 - April 2024

- Constructed a convolutional neural network (CNN) in Keras to classify EMNIST letter images, achieving improved accuracy over an equivalent dense feedforward model
- Designed and trained a generative adversarial network (GAN) in PyTorch to produce synthetic EMNIST letter images, enhancing dataset diversity and supporting data augmentation strategies
- Evaluated model performance with visual inspection and quantitative similarity metrics between generated and real samples

SKILLS

Programming: C++ (proficient), Python (proficient), MySQL (proficient), Java (beginner), MATLAB (beginner), JavaScript (beginner), HTML (beginner), CSS (beginner), CUDA (beginner), Solidity (beginner)

Machine Learning: Keras (proficient), PyTorch (beginner), Scikit-Learn (proficient), Pandas (proficient), NumPy (proficient), Tensorflow (beginner)

Software/Hardware: Microsoft Office (proficient), Linux (proficient), Simulink (beginner), SPICE (beginner), Logisim (beginner), LabVIEW (beginner), Unity (beginner), Arduino (beginner), Verilog (beginner), Raspberry Pi (beginner)

Languages: English (fluent), Persian (fluent), Arabic (basic understanding), Spanish (basic understanding)